

Target's Pregnancy Prediction

Predictive Analytics & Privacy Ethics

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Algorithm Basics and BW

Target's Pregnancy Prediction

Predictive Analytics & Privacy Ethics

When retail analytics crosses the privacy line

Learning Objectives

1.  How predictive analytics uses purchase patterns to infer personal information
2.  Business value of behavioral targeting and customer lifetime value
3.  Privacy implications of predicting life events without consent
4.  The “creep factor” and how to manage prediction transparency
5.  GDPR/CCPA compliance considerations for predictive models

Key Question: Where's the line between personalization and invasion of privacy?

Background: Retail Analytics in the 2000s

Traditional marketing:

- ▶  Mass mailings
- ▶  Newspaper inserts
- ▶  TV commercials
- ▶  Low response rates (1-2%)
- ▶  Expensive waste
- ▶  No personalization

The opportunity:

- ▶ Transaction data available
- ▶ Loyalty cards track purchases
- ▶ Can we predict behavior?

Background: Why Target Pregnant Women?

Why target pregnant women?

- ▶  New parents = high-value customers
- ▶  Spend 60% more
- ▶  Shopping habits change
- ▶  Brand switching opportunity
- ▶  Early acquisition = lifetime loyalty

The challenge:

- ▶ Women don't announce pregnancy
- ▶ Baby registries only in 2nd trimester
- ▶ Need to identify EARLY

Target's Data Infrastructure

Data collection:

- ▶  Guest ID system
 - ▶ Tracks every purchase
 - ▶ Links credit card, email, phone
- ▶  Point-of-sale transactions
- ▶  Web browsing behavior
- ▶  Email engagement
- ▶  Baby registry data
- ▶  Third-party data purchases

What they knew:

- ▶  Demographics (age, location)
- ▶  Purchase history
- ▶  Shopping frequency
- ▶  Spending patterns
- ▶  Seasonal trends
- ▶  Store visit patterns

Key insight: Every transaction reveals something about life circumstances

The Pregnancy Prediction Project

Project leader:

- ▶  **Andrew Pole**, Senior Statistician
- ▶  Goal: Predict pregnancy
- ▶  Timeline: Early 2000s
- ▶  Using machine learning

The hypothesis:

- ▶ Pregnant women buy specific products
- ▶ Purchase patterns change by trimester
- ▶ Combinations more predictive than individual items
- ▶ Can identify pregnancy early (1st trimester)

The Pregnancy Prediction Project

Training data source:

- ▶ 📺 Baby registry customers
 - ▶ Known pregnant
 - ▶ Know due dates
 - ▶ Can look backward
- ▶ 🗄 Thousands of examples
- ▶ ✅ High-quality labels

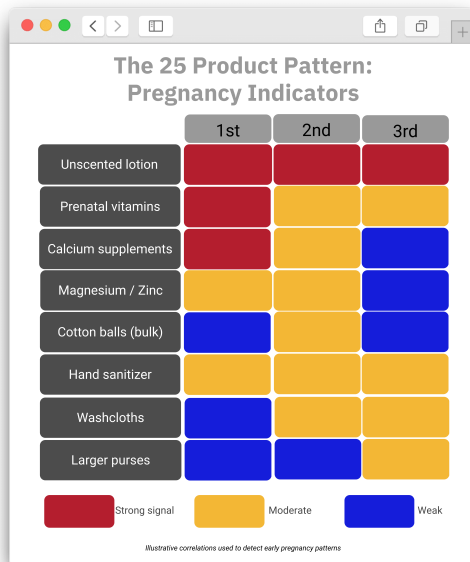
The approach:

- ▶ Supervised machine learning
- ▶ Identify product patterns
- ▶ Test on new customers
- ▶ Refine model iteratively

The 25 Product Pattern

Key products identified:

- ▶  Unscented lotion
 - ▶ Scent sensitivity in pregnancy
- ▶  Supplements
 - ▶ Calcium, magnesium, zinc
 - ▶ Prenatal vitamins
- ▶  Cotton balls (large quantities)
- ▶  Hand sanitizer
- ▶  Washcloths
- ▶  Larger purses



Predictive Model Development - Parameterization

Predictive Score for pregnancy from model:

$$z = \alpha$$

+ β_1 Unscented Lotion (Yes = 1 / No = 0)

+ β_2 Prenatal Vitamins (Yes = 1 / No = 0)

+ β_3 Calcium Supplements (Yes = 1 / No = 0)

+ ϵ

$$Prob(\text{Pregnancy}) = \frac{1}{1 + e^{-z}}$$

Beyond Individual Products: Pattern Recognition (Why combinations matter)

Single products:

- ▶ ❌ Low predictive power
- ▶ ❓ Many false positives
- ▶ Example: Lotion purchase
 - ▶ Could be: pregnant, dry skin, gift
 - ▶ Not enough information

Product combinations:

- ▶ ✔ High predictive power
- ▶ Example: Unscented lotion + calcium + zinc + cotton balls
- ▶ Probability: 87% pregnant

Predictive Model Development

Predictive Score for pregnancy from model based on combo:

$$z = \alpha$$

+ β_1 Unscented Lotion (Yes = 1 / No = 0)

+ β_2 Prenatal Vitamins (Yes = 1 / No = 0)

+ β_3 Calcium Supplements (Yes = 1 / No = 0)

+ β_4 Unscented Lotion & Prenatal Vitamins (Yes = 1 / No = 0)

+ β_5 Prenatal Vitamins & Calcium Supplements (Yes = 1 / No = 0)

+ β_6 Calcium Supplements & Unscented Lotion (Yes = 1 / No = 0)

+ β_7 All 3 items (Yes = 1 / No = 0)

+ ϵ

$$Prob(\text{Pregnancy}) = \frac{1}{1 + e^{-z}}$$

Beyond Individual Products: Pattern Recognition (Why combinations matter)

Timing patterns:

- ▶ 📅 Purchase sequence matters
- ▶ 🕒 Frequency changes
- ▶ Week 1-12: Supplements
- ▶ Week 13-26: Maternity clothes
- ▶ Week 27-40: Baby items

Prediction accuracy:

- ▶ Can predict due date within small window
- ▶ Can identify trimester
- ▶ 25 products analyzed together
- ▶ Sophisticated scoring algorithm

Key insight: Context + patterns + timing = powerful predictions

Business Value: How the revenue model works: \$600M Annually

Customer lifetime value:

- ▶  Average customer: \$50/month
- ▶  New parent: \$80/month (60% increase)
- ▶  Duration: 18+ years
- ▶  Lifetime value:
 - ▶ $\$80 \times 12 \times 18 = \$17,280$
- ▶  vs regular customer: \$10,800
- ▶  Difference: \$6,480 per parent

Key driver: Acquiring customers BEFORE competitors do

The \$600M calculation:

- ▶  4M births/year in US
- ▶  Target captures 2.5% = 100K
- ▶  Early targeting = 30% higher retention
- ▶  30K additional customers $\times$ \$6,480
- ▶  $\rightarrow$ = \$194M in lifetime value
- ▶  Plus immediate pregnancy spending
- ▶  Total: \$600M+ annually

Marketing Efficiency Gains

Traditional marketing:

- ▶  Mass mailing to everyone
- ▶  \$1 per mailer
- ▶  1-2% response rate
- ▶  Cost per acquisition:
 - ▶ $\$1 / 0.015 = \67
- ▶  98% waste

Targeted marketing:

- ▶ Mail only to high-probability pregnant women
- ▶ 25% response rate
- ▶ $\$1 / 0.25 = \4 cost per acquisition

The Famous Story: Target Knew First (The incident that made headlines)

What happened:

- ▶ 📧 Target mails baby coupons
- ▶ 🏠 To teenage girl's home
- ▶ 😡 Father sees coupons, complains
"She's still in high school!"
- ▶ 🛒 Store manager apologizes
- ▶ ⌚ Few weeks later...
- ▶ 😞 Father apologizes to Target
 - ▶ Daughter was pregnant
 - ▶ She hadn't told him
- ▶ **Target knew before he did**

Target's Response: The Camouflage Strategy

The problem:

- ▶  Customers felt “spied on”
- ▶  Backlash hurting brand
- ▶  But predictions were valuable
- ▶  How to use data without creeping people out?

The solution:

- ▶  Camouflage baby products
- ▶  Mix with random items
- ▶ Example mailer:
 - ▶ Diapers + wine glasses
 - ▶ Baby lotion + lawn mower
 - ▶ Crib sheets + power tools

Target's Response: The Camouflage Strategy

The psychology:

- ▶  Looks like general catalog
- ▶  Pregnant women notice baby items
- ▶  Others ignore them
- ▶  Reduces “creep factor”
- ▶  But is this more ethical?

The debate:

- ▶  More honest or more deceptive?
- ▶  Pro: Respects privacy perception
- ▶  Con: Hides surveillance
- ▶  Which is worse?

Question: Is hiding your prediction capabilities ethical?

Lesson 1: Prediction Accuracy $\neq$ Ethical Use

Technical achievement:

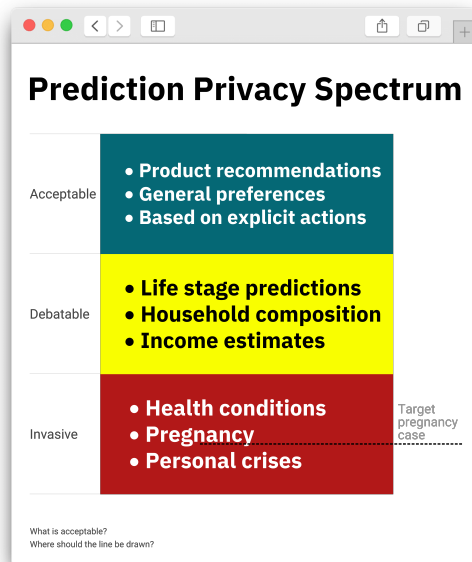
- ▶  Remarkable ML success
- ▶  High prediction accuracy
- ▶  Business value clear
- ▶  Sophisticated analytics
- ▶  Measurable ROI

But ethical questions:

- ▶  Just because we CAN predict...
- ▶  Should we?
- ▶  Who consents?
- ▶  What are boundaries?

Privacy Spectrum: Where to draw the line

- ▶ ✓ **Acceptable**
- ▶ ? **Debatable**
- ▶ ✗ **Invasive**



Lesson 2: Training Data Quality Matters

Why Target's approach worked:

- ▶ 📺 Baby registry = gold standard
 - ▶ Known pregnancy status
 - ▶ Known due dates
 - ▶ High-quality labels
- ▶ 🗄 Thousands of examples
- ▶ ✅ Can look backward
 - ▶ See what they bought before registry
- ▶ 🧠 Pattern detection possible

Potential problems:

- ▶ ⚠ Selection bias
 - ▶ Only women who register
 - ▶ May miss patterns
- ▶ ❓ False positives
 - ▶ Gift buyers
 - ▶ Helping friends
- ▶ ✖ False negatives
 - ▶ Different shopping patterns
 - ▶ Online vs in-store

Lesson 3: Transparency vs Effectiveness (The tension)

Full camouflage:

- ▶ ✓ Maintains effectiveness
- ▶ ✓ No backlash
- ▶ ✓ Business value preserved
- ▶ ✗ Deceptive
- ▶ ✗ Hides surveillance
- ▶ ✗ Erodes autonomy

Example:

- ▶ Mix baby items with random products
- ▶ Pretend it's general catalog
- ▶ Hide the prediction

Question: Is there a middle ground? Should companies be required to disclose?

Would Target's approach be legal today?

GDPR requirements:

▶ Lawful basis:

- ▶ Legitimate interest?
- ▶ Or explicit consent needed?

▶ Transparency:

- ▶ Right to know what data used
- ▶ Right to know how processed

▶ Purpose limitation:

- ▶ Can't use purchase data for unrelated purposes

▶ Data minimization:

- ▶ Collect only what's needed

Ethics Discussion: Drawing the Line

1.  Should there be a legal distinction between predicting:
 - ▶ Shopping preferences (“likes blue shirts”) vs
 - ▶ Personal life circumstances (“is pregnant”)?

If yes, where exactly is the line?

2.  Target's camouflage strategy:
 - ▶ More ethical (respects privacy perception)?
 - ▶ Less ethical (hides surveillance)?
 - ▶ Or just better PR?

Ethics Discussion: Drawing the Line

3.  Who owns insights derived from your data?
 - ▶ You own transaction history
 - ▶ Does Target own the pregnancy prediction?

4.  What would meaningful consent look like?
 - ▶ “We may infer pregnancy from purchases”?
 - ▶ Too specific to be practical?
 - ▶ Too vague to be meaningful?

Ethics Discussion: Consent & Control

5. 💰 Business vs privacy trade-off:

- ▶ Target gains \$600M/year
- ▶ Customers lose privacy
- ▶ Fair exchange?
- ▶ Should customers be compensated?

6. 🔑 Should regulations require:

- ▶ Disclosure of predictive models?
- ▶ Opt-out mechanisms?
- ▶ Accuracy audits?
- ▶ All of the above?

Modern Implications: This isn't just Target

Similar predictions today:

- ▶  Facebook: Relationship status
- ▶  Google: Income level
- ▶  Amazon: Financial stress
- ▶  Apple: Health conditions
- ▶  Insurance: Driving risk
- ▶  Universities: Dropout risk

The pattern:

- ▶ Behavioral data reveals everything
- ▶ Predictions getting more accurate
- ▶ Boundaries still unclear

Modern Implications

Emerging concerns:

- ▶  Health condition predictions
 - ▶ Insurance discrimination
- ▶  Financial distress
 - ▶ Predatory lending
- ▶  Academic performance
 - ▶ Admissions bias
- ▶  Relationship stability
 - ▶ Employment discrimination

Question: As predictions get better, do we need stronger protections?

Key Takeaways

1.  **Transaction data is personal data:** Every purchase reveals life circumstances
2.  **Patterns beat individual data points:** Combinations are powerfully predictive
3. **\$ Business value is real:** \$600M annually from one predictive model
4.  **Accuracy $\neq$ Ethics:** Just because we can predict doesn't mean we should
5.  **“Creep factor” matters:** Even accurate predictions can backfire
6.  **Regulations catching up:** GDPR/CCPA now restrict these practices
7.  **Lines are blurry:** Where's the boundary between personalization and invasion?

References

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